Introduction to JavaScript

What can JavaScript Do?

- JavaScript can dynamically modify an HTML page
- JavaScript can react to user input
- JavaScript can validate user input
- JavaScript is a full-featured programming language
- JavaScript user interaction does not require any communication with the server

Pros: – Allows more dynamic HTML pages, even complete web applications

Cons: – Requires a JavaScript-enabled browser – Requires a client who trusts the server enough to run the code the server provides

Where to Put your Scripts

Scripts can be placed in the HEAD or in the BODY
– In the HEAD, scripts are run before the page is displayed
– In the BODY, scripts are run as the page is displayed

```
<html>
<head>
<title> JS </title>
</head>
<body>
   <script type = "text/javascript">
      document.write("Hello World!");
   </script>
</body>
</html>
```

External js uses '.js' file
<script src="app.js"></script>

Javascript Variables

JavaScript has variables that you can declare with the **let** keyword

let name = ‘Rakib’;
let i = 1.4;
let j = 2;

Data Types:

- Number
- String
- Boolean
- Undefined
- Null
- Object
- Array

JavaScript has most of the operators we're used to from C/Java
Arithmetic (+, - ,*, /, %)
Assignment (=, +=, -=,*=, /=, %=, ++, --)
Logical (&&, ||, !)
Comparison (<, >, <=, >=,==)

It has basic **Control Statements:**

- if, else, else if
- switch

```
if( i== '15')  {
    console.log("You are logged in");
} else {
    console.log("Wrong number");
}
```

Loops:

- for
- while
- do….while

```
for ( let x = 0; x <=10; x++){
        console.log(x);
    }
```

Simple User Interaction

There are three built-in methods of doing simple user interaction

- **alert**(msg) alerts the user that something has happened
- **confirm**(msg) asks the user to confirm (or cancel) something
- **prompt**(msg, default) asks the user to enter some text

```
let i = prompt("Enter your number: ");
i = Number(i);
let j = 2;
let z = i + j;

alert(“The value of z is” + z);

confirm(“Do you want to proceed”);
```

Prompt takes the value as string by default. We need to typecast it to number if we want to arithmetic operation.

JavaScript Functions

JavaScript lets you define functions using the **function** keyword.
You dont need to specify the return type.

```
function printTen(){
        for(let x = 0; x <=10; x++){
            console.log(x);
        }
    }
```

You can return value too using return keyword;

```
function add(a, b) {
  return a + b;
}
```

JavaScript Arrays

- Store multiple values in one variable
- Zero-indexed
- Can store different type of values in a single array
    - let mixed = [“BD”, 1, 1.5];

let fruits = ["Apple", "Banana", "Mango"];
Access by using index.
i.e: fruits[0] returns “Apple”

fruits.length will give us the size of the array.

To iterate through the array:

```
for(let item of fruits){
        console.log(item);
    }
```

Array methods:

push(value)	Adds element at the end
pop()	Removes last element
unshift(value)	Adds element at the beginning
shift()	Removes first element

splice(index to start from, number of elements to remove)
 	
`let fruits = ["Apple", "Banana", "Mango", "Orange"];`
 	
`fruits.splice(2, 1);`
 	
It removes item at index 2.
 	
splice(index to start from, number of elements to remove, element to add)
 	
`let fruits = ["Apple", "Banana", "Orange"];`
`fruits.splice(1, 0, "Mango");`
	Removes elements from a specific index.
	
	Add element at a specific index.

This adds "Mango" at index 1	
sort()	Sorts the array
reverse()	Reverses array

These functions modify the actual array.
More methods at https://www.w3schools.com/Js/js_array_methods.asp

`map()`

- Goes through **each element** of an array
- Applies a **function** to each element
- Returns a **new array**
- **Does NOT change** the original array

```
let arr2 = [5, 10, 15, 20, 25];

let result= arr2.map(function double(x){
        return x*2;
    });
```

Shorthand arrow (=>) Notation

We can use arrow function (lambda function) instead of declaring the whole function.

```
let result= arr2.map ( x => x*2 );
```

```
let fruits = ["apple", "banana", "mango"];

let upperFruits = fruits.map(fruit => fruit.toUpperCase());
```

Be aware of this common mistake

```
numbers.map(num => {
  num * 2;
});
//result will be [undefined, undefined, ...]
numbers.map(num => {
  return num * 2;
});
```

```
// This is correct
```

If you use a second bracket you will have to specifically write **return** else the result will be undefined.

`filter()`

- Goes through **each element** of an array
- Checks a **condition**
- Returns a **new array** with only the elements that satisfy the condition
  Does NOT change the original array

```
function isEven(num) {
  return num % 2 === 0;
}

let numbers = [1, 2, 3, 4, 5, 6];

let evens = numbers.filter(isEven);
```

```
let words = ["apple", "banana", "kiwi", "orange"];

let longWords = words.filter(word => word.length > 5);
```

```
arr2 = [5, 10, 15, 20, 25];
blacklist = [10, 20];

let filtered = arr2.filter(x => !blacklist.includes(x));
```

`some()`:

- Checks if **at least one element** satisfies a condition
- Returns **true or false**

```
let numbers = [2, 4, 6, 7];

let hasOdd = numbers.some(num => num % 2 !== 0);

//returns true.
```

`every()`

- Checks if **all elements** satisfy a condition
- Returns **true or false**

```
let numbers = [2, 4, 6, 8, 9, 11];

let allEven = numbers.every(num => num % 2 === 0);
// Returns false
```

`forEach()`

- Executes a function for **each element**
- Does **not return anything**
- Usefull to iterate the array

```
numbers.forEach( value => {
   console.log(value);
})
```

`join()`

- Combines all array elements into a **single string**
- Uses a **separator** between elements
- Does **not change** the original array

```
let fruits = ["Apple", "Banana", "Mango"];

let result = fruits.join(", ");

//result will be “Apple, Banana, Mango”
```

JavaScripts String Methods

length	Returns the length of the string
charAt(index)	Returns the character at a given index
toUpperCase()	Converts string to uppercase
toLowerCase()	Converts string to lowercase
slice(start, end)	Extracts part of a string
replace(old, new)	Replaces first occurrence
replaceAll(old, new)	Replaces all occurrences
trim() **trimStart()** **trimEnd()**	Removes spaces from both ends Removes spaces from start Removes spaces from end
includes(value)	Checks if string contains value
split(separator)	Converts string to array of substring
startsWith(value) **endsWith(value)**	Checks if string starts with value Checks if string ends with value
indexOf(value) **lastIndexOf(value)**	Returns first index of value Returns last index of value
repeat(count)	Repeats the string

All JavaScript string methods return a new string. They do NOT modify the original string.

Find a lot more in https://www.w3schools.com/js/js_string_methods.asp

String Practice Problems

1. Write a JavaScript function to split a string and convert it into an array of words!
2. Write a JavaScript function to extract first four characters from a string.
3. Write a JavaScript function to convert a string in abbreviated form. (Mahmudul Hasan - M.H)
4. Write a JavaScript function to hide email addresses to protect from unauthorized user. (johnclerk@gmail.com » john*****@gmail.com)
5. Write a JavaScript function to put hyphen between the words of a string. (Bangladesh is a country + Bangladesh-is-a-country)
6. Write a JavaScript function to capitalize the first letter of each words of a string
7. Write a JavaScript function that takes a string which has lower and upper case letters as a parameter and converts upper case letters to lower case, and lower case letters to upper case
8. Write a JavaScript function to concatchates a given string n times (default is 1).

https://www.w3resource.com/javascript-exercises/javascript-string-exercises.php

Array Practice Problems:

1. Write a JavaScript function to get the last element of an array.
2. Write a simple JavaScript program to join all elements of the following array into a string.
3. Write a JavaScript program to sort the items of an array in descending order.
4. Write a JavaScript program to find the sum of all the items.
5. Write a JavaScript program to double all the elements in an array.
6. Write a JavaScript program to double all the elements in an array.
7. Write a JavaScript program to find all the elements greater than 5 in an array.
8. Write a JavaScript program to remove duplicate items from an array

https://www.w3resource.com/javascript-exercises/javascript-array-exercises.php